

Instrument: Green Climate Fund / Adaptation Fund screening note (auto-filled)
Status: model output ? screening-grade optimisation, not an investment recommendation

#1. Problem

Monsoon floods, rainfall-triggered landslides, and glacial-lake outburst risk concentrate below Tsho Rolpa in the Dudh Koshi. NOAA ISD annual maxima (GEV) show the early-period 1-in-100-year daily rainfall depth now has a fitted recurrence of 7.75 years in the late-period sample ([1979, 1999] vs [2000, 2024]). Trend in annual maxima: 5.229 mm/decade.

Observed lake-area change 1990?2024: Imja Tsho +113.3%, Thulagi +30.3%, Tsho Rolpa +26.6%.

#2. Evidence

- Stations: 498 / station-years: 12066 (NOAA ISD).
- Return levels (mm): {'2': 29.359, '5': 41.664, '10': 49.032, '25': 57.54, '50': 63.321, '100': 68.648} with bootstrapped 95% CIs.
- Landslide: rainfall classifier, AUC out-of-sample ? not a Caine-style I?D threshold; AUC 0.934 on 138 events, test [2016, 2017].
- Backtest: CSI 0.086 (event 2024-09-27)
- Counterfactual: Counterfactual exposure 54993.4 -> 26845.7 (51.18% reduction; observed-flood xpopulation, not unique lives).
- POT/GPD cross-check return levels (mm): {'2': 230.253, '5': 282.957, '10': 324.123, '25': 380.314, '50': 424.203, '100': 469.32}
- IMERG: Daily IMERG is not on an unauthenticated public URL. Use Google Earth Engine NASA/GPM_L3/IMERG_V07 or Earthdata if credentials exist. RootLedger's headline remains NOAA ISD GEV.
- Hazard layer: D8 HAND on Copernicus GLO-30, stage calibrated to UNOSAT S-1 27 Sep 2024; screening-grade NumPy, not Whitebox.

#3. Intervention

Portfolio of nature-based parcels (vetiver, bamboo, afforestation, floodplain/wetland restore, riverbank bioengineering) selected by marginal greedy triple-return per dollar with per-cell EAL capping and Monte-Carlo climate noise from the GEV 100-year CI.

#4. Budget and expected impact

Item Value
--- ---
Budget (USD) 2,000,000
Mode expected
Parcels selected 127
Spend (USD) 1,998,560
Annual expected people-risk avoided 67944.2
CO? sequestered, 10 yr (t) 17,554
Household income (USD/yr) 226,288
Households benefiting 915,128

'people_protected' is annual expected people-risk avoided, not a count of unique lives.

#5. M&E

Satellite MRV: annual NDWI lake area, Sentinel-1 flood extent after events, parcel presence via high-res optical. Re-run EVT as NOAA ISD updates.

#6. Citations (screening-grade factors)

See 'plan.provenance.factors' for per-type cost, efficacy, carbon, and income sources.